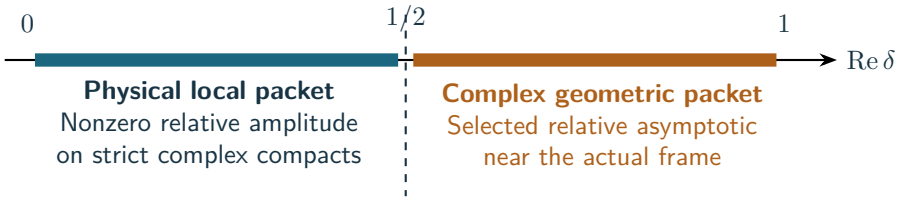
One exceptional edge at any fixed bridge width



Folded limiting channel: $a/N \to 1/2$, $\delta = h/N$.
At every finite level, physical entries with $h > a$ vanish exactly.

A finite formula across the support wall

$$\mathcal{K}_{N,a} = \frac{\beta(2a,a)^2}{N} \prod_{r=a+1}^{N-1-a} 2\sin\frac{\pi(h+r)}{N} R_a(z_1;c) R_a(z_2;c)$$

Apparent denominators and phases cancel exactly.
The entire root kernel retains all physical lattice zeros.

Still needed for the general theorem

A nonzero geometric contribution to the original global period,
with boundary residues and competing colors controlled.